

Liam Dubai

Astrophysics PhD Candidate
The Ohio State University

dubay.11@osu.edu | www.liamdubay.com

Education

The Ohio State University

PhD Candidate in Astronomy and Astrophysics

Columbus, OH

2021 – Present

Whitman College

B.A. in Physics-Astronomy, Music (Performance), *summa cum laude*

Walla Walla, WA

May 2021

Honors in Physics-Astronomy, Thesis Advisor: Prof. Andrea K. Dobson

Late-Onset Circumstellar Medium Interactions are Rare: An Unbiased GALEX View of Type Ia Supernovae

Honors in Music (Performance), Thesis Advisor: Dr. Sally Singer Tuttle

Honors & Awards

NSF Graduate Research Fellowship Proposal, Honorable Mention

2022

The Ohio State University Graduate Fellowship

2021 – 2022

Financial support for first-year graduate students

Campbell Music Scholarship

2017 – 2021

Recognition of exceptional musical talent and achievement

Walter Brattain Scholarship

2017

High academic excellence in college preparatory work

William O. Douglas Valedictorian Scholarship

2017

Research

Publications

Dubay, L., Johnson, J.A., & Johnson, J.W. (2024), “Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution”, *ApJ* 973, 55, doi:10.3847/1538-4357/ad61df

Dubay, L. et al. (2022), “Late-Onset Circumstellar Medium Interactions are Rare: An Unbiased *GALEX* View of Type Ia Supernovae”, *ApJ* 926, 98, doi:10.3847/1538-4357/ac3bb4

Conference Presentations

- SDSS-V Collaboration Meeting**, Las Cruces, NM 2024
Talk: “Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution”
- Surveying the Milky Way: The Universe in Our Own Backyard** 2023
California Institute of Technology, Pasadena, CA
Poster: “The Galactic Delay-Time Distribution of Type Ia Supernovae: A Chemical Evolution Perspective”
- 241st American Astronomical Society Meeting**, Seattle, WA 2023
Talk: “The Delay Times of Type Ia Supernovae: A Chemical Evolution Perspective”
- 237th American Astronomical Society Meeting**, Virtual 2021
Poster: “Late-Onset Circumstellar Medium Interactions are Rare: An Unbiased *GALEX* View of Type Ia Supernovae”
- 235th American Astronomical Society Meeting**, Honolulu, HI 2020
Poster: “Investigating the stationarity of sensing noise in LISA Pathfinder data”

Undergraduate Research Experience

- Institute for Astronomy REU**, University of Hawai‘i 2020
Advisor: Dr. Benjamin J. Shappee
- Heliophysics REU**, The University of Alabama in Huntsville / 2019
NASA Marshall Space Flight Center
Advisor: Dr. Tyson Littenberg
- Earle A. Chiles Research Institute Summer Research Program** 2017
Providence Cancer Institute, Portland, OR
Advisor: Dr. Brady Bernard

Teaching

- Graduate Teaching Assistant**, The Ohio State University 2022 – 2023
Astronomy 3350: Methods of Astronomical Observation & Data Analysis Fall 2023
Astronomy 1221: Astronomy Data Analysis Spring – Fall 2023
Astronomy 1101: Planets to Cosmos Lab (2 sections) Fall 2022
- Astronomy Teaching Assistant & Tutor**, Whitman College 2019 – 2020
Astronomy 177: Sky and Planets Spring 2020
Astronomy 179: Galaxies and Cosmology Fall 2019
Astronomy 178: Sun and Stars Spring 2019

Outreach

- Planetarium Presenter**, Arne Slettebak Planetarium 2021 – Present
The Ohio State University, Columbus, OH
Audiences include: K-12, boy & girl scouts, university groups, and the general public.
- Library Talks**, Columbus Metropolitan Libraries 2023 – Present

Service & Mentoring

- “Galaxy Hour” Meeting Co-Organizer**, Ohio State Astronomy Dept. 2023 – Present
- Polaris Leadership Committee**, The Ohio State University 2023 – Present
Website: <https://u.osu.edu/polaris>
Graduate student-led organization dedicated to fostering a more inclusive environment and improving retention of underrepresented minority groups in physics and astronomy.
- Mentor**, Polaris Mentorship Course 2022 – Present
Graduate mentors are paired with first- or second-year undergraduates from underrepresented backgrounds during this year-long course, which includes a small research project and poster presentation.